Automotive Engineering 🔧

Junior Automotive Engineering Badge — Girls Build Bots

Presented by your robotics mentors!

Today we test and measure 

Design it → **engineer it to perform** → manufacture it.

This is the **test-and-measure** badge: real trials, real data, and a passenger to protect.

Propulsion: the balloon engine 

Blow up a balloon. Let it go.

Why does it fly?

- Air pushes out **backward**
- The balloon goes **forward**
- That's **thrust** — action and reaction

Today’s engine is a balloon 

The engineering isn’t the balloon.

It’s making that thrust move a car straight and far.

Build the balloon car 🛠️

Your team builds: - A **chassis** - **Free-spinning** axles - **Wheels** - A balloon with a **straw nozzle** taped on

First milestone: it moves under its own power.

Classic failures (and they’re all fixable) 🔍

- Wheels rubbing the chassis → **friction**
- Nozzle pinched → no thrust
- Car too heavy → goes nowhere

Ask yourself: where is it losing energy?

Test Trials — with DATA 📋

Three timed rounds. Every round:

1. **Run** it
2. **Measure** the distance 📏
3. **Record** it on the data sheet
4. Change **exactly one variable**
5. **Predict** the effect before re-running

Engineers don’t remember — they RECORD 

Trial	What we changed	Distance	Notes
1	—		
2			
3			

No trial counts unless it’s written down.

Change three things at once? Then which one worked? 🤷

Crash Safety Challenge 

New requirement: your car carries an egg passenger down the ramp into the wall…

…and the passenger must SURVIVE.

Engineer restraints and crumple zones. Then crash-test. Redesign once if your passenger gets hurt.

Real automotive safety 

The vehicle can fail.

The passenger can’t.

Robot moment 

Our team runs test matches and **logs data** on the robot, changing one thing between tests.

Same trials. Bigger machine.

Wrap 🎯

Share: 1. **One number** from your data sheet 2. **One thing** that surprised you

What you accomplished ✅

- Explained how thrust propels a balloon car (action–reaction)
- Built a vehicle that moves under its own power
- Ran fair tests — one variable at a time — and recorded the data
- Engineered passenger protection and crash-tested it

Keep going at home! 

- Rebuild your car and beat your best recorded distance
- Run a fair test on anything: change ONE thing, measure, repeat
- Ask your troop leader about the next Automotive badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.